

Context-Free Grammars (CFG)

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Not all languages are regular. In fact, there are uncountable many languages over any alphabet Σ , but only countably many of them are regular. Therefore, most languages are not regular.

The first lets consider some characterization for regular languages which we will use to prove that some languages are not regular.

Recall, that language L is regular if and only if there is DFA which accept the language L .

Pumping Lemma

Lemma

If a language L is accepted by a DFA M with n states, then every string $x \in L$ with $|\omega| \geq n$ can be written as $\omega = xyz$ such that $y \neq \varepsilon$ and $xy^kz \in L$.

Prove in whiteboard

Pumping Lemma as Quantified Formula

If L is regular, then

$$\begin{aligned} \exists n \in \mathbb{N} \quad \forall \omega \in L \quad \exists x, y, z \quad \forall k \quad [xy^kz \in L] \\ |\omega| \geq n \quad \omega = xyz \\ y \neq \varepsilon \end{aligned}$$

Recall that the following statements are equivalent for any statements A and B :

- $A \rightarrow B$;
- $\neg B \rightarrow \neg A$.

Equivalent formula for Pumping lemma

If

$$\begin{array}{llll} \forall n \in N & \exists \omega \in L & \forall x, y, z & \exists k \quad [xy^kz \notin L] \\ & |\omega| \geq n & \omega = xyz & \\ & & y \neq \varepsilon & \end{array}$$

then L is not regular.

Examples

Prove that the following languages are not regular:

- 1 $L = \{0^n 1^n : n \in N\};$
- 2 $L = \{0^n 1^m : n \neq m\};$
- 3 $L = \{\omega \in \{0, 1\}^* : \omega = \omega^R\};$
- 4 $L = \{\omega \in \{0, 1\}^* : \#_0(\omega) \neq \#_1(\omega)\};$
- 5 $L = \{0^p : p \text{ is prime}\}.$

Context Free Grammars

In study of natural languages, a grammar is a set of rules that govern how the sentences in a language are generated. For instance, in English grammar contains the following rules:

$$\begin{aligned}\langle \textit{Sentence} \rangle &\rightarrow \langle \textit{Subject} \rangle \langle \textit{Predicate} \rangle \\ \langle \textit{Subject} \rangle &\rightarrow \langle \textit{Noun} \rangle \\ \langle \textit{Predicate} \rangle &\rightarrow \langle \textit{Verb} \rangle \langle \textit{Adverb} \rangle \\ \langle \textit{Noun} \rangle &\rightarrow \textbf{water} \\ \langle \textit{Verb} \rangle &\rightarrow \textbf{vaporates} \\ \langle \textit{Adverb} \rangle &\rightarrow \textbf{constantly}\end{aligned}$$

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To explain how these grammar rules work, let's call terms of the form $\langle \dots \rangle$ **nonterminal symbols** and words in English **terminal symbols**.

We call a string formed by nonterminal and terminal symbols a **sentential form**, and call a string formed by terminal symbols only a sentence. We can generate a sentence from the previous grammar rules as follows:

- 1 First, we create a sentential form of a single symbol $\langle \textit{Sentence} \rangle$;
- 2 We repeatedly perform the substitution of the grammar rules until the sentential form contains only terminal symbols:
 - We select a nonterminal symbol $\langle a \rangle$ in the sentential form and replace it by the right-hand side of the grammar rule whose left-hand side is $\langle a \rangle$.

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We call a grammar with rules of this form a **context-free grammar**

Formally, we define a **context-free-grammar** (**CFG** in short) to be a quadruple (V, Σ, R, S) of the following four components:

- V is a finite set of non-terminal symbols;
- Σ is a finite set of terminal symbols;
- R is a finite set of rules which are of the form $A \rightarrow x$, where $A \in V$ and $x \in (V \cup \Sigma)^*$;
- S is a special symbol in V , called a **starting symbol**.

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Let $G = (V, \Sigma, R, S)$ be a CFG. Then, we call a string $x \in (V \cup \Sigma)^*$ a **sentential form**, and a string $y \in \Sigma^*$ a **sentence**.

Let u and v be strings in $(V \cup \Sigma)^*$ and $A \in V$. Then we write

$$uAv \xrightarrow[G]{} uxv$$

if $(A \rightarrow x)$ is a rule in R . That is, we can substitute string x for symbol A in a sentential form if $(A \rightarrow x)$ is a rule in R .

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For any two sentential form x and y , we write $x \xrightarrow[G]{*} y$, if there are sequences $x = x_0, x_1, x_2, \dots, x_n = y$ of sentential forms such that $x_i \xrightarrow[G]{} x_{i+1}$ for all $i < n$.

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If $S \xrightarrow[G]^* y$ for $y \in \Sigma^*$, then we say the CFG G **generates** the sentence y , and say the sequence

$$S = x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_n = y$$

is a **derivation** of the sentence y .

The language $L(G)$ generated by G is the set of all strings over Σ that are generated by G , that is

$$L(G) = \{x \in \Sigma^* : S \xrightarrow[G]{*} x\}.$$

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Example. Consider the CFG $G = (\{S\}, \{0, 1\}, R, S)$ where R consists of the following rules

$$S \rightarrow \varepsilon, \quad S \rightarrow 0S1.$$

What is the language $L(G)$?

Examples.

Find context-free grammars that generates the following languages:

- 1 $L = \{0^n 1^{2n} : n \geq 0\};$
- 2 $L = \{\omega \in \{0, 1\}^* : \omega = \omega^R\};$
- 3 $L = \{\omega \in \{0, 1\}^* : \#_0(\omega) = \#_1(\omega)\};$
- 4 $L = \{\omega \in \{a, b\}^* : \#_a(\omega) = 2\#_b(\omega)\};$
- 5 $L = \{a^m b^n c^p d^q : m + n = p + q\}.$

Theorem

For each regular language L , there exists a CFG G such that $L = L(G)$.

Proof on whiteboard ...

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Note that context-free languages are also countable, that is there are languages that are not context-free.